

Cumulative Volcanic Burial and the Archaeological Detection Horizon in Java, Indonesia: Preliminary Measurement Anchors and a Research Program

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Abstract

Java hosts 45 active volcanoes in 129,000 km², and its archaeological record becomes progressively sparser with age in the volcanic interior. We ask a narrowly geomorphological question: at what rate has cumulative volcanic sedimentation buried the landscape, and what does that rate imply for the detection horizon of surface survey? We present preliminary measurement anchors and propose a research program to test them rigorously.

In 1803, the Dutch colonial administrator Nicolaus Engelhard documented two stone guardian statues at Candi Singosari in East Java—each 370 cm tall, carved from monolithic andesite in 1268 CE—with approximately half their bodies below ground. Gunung Kelud had erupted about twenty times in the intervening 535 years, and the landscape had risen around them at approximately 3.5 mm per year. The archival precision of this colonial-era observation requires primary-source verification, and we treat it here as a preliminary anchor rather than a validated calibration point. We extend the inference with three additional measurement anchors from the Merapi system in Central Java (Sambisari, Kedulan, Kimpulan, all 9th century CE), obtaining site-level rates of 2.4–6.2 mm/yr across four sites in two volcanic systems and 350 km of Java. With $n = 4$ and a shared methodological basis (depth divided

by elapsed time since dated construction), these site-level rates fall within an order of magnitude but do not yet constitute a validated cross-regional calibration. We present them as preliminary measurement anchors requiring further geoarchaeological testing before they can be generalised.

Applied as a first-order linear projection at the site-level mean, these anchors place Kanjuruhan-era (~760 CE) deposits at 3.0–7.9 m and pre-400 CE deposits at 3.9–10.1 m depth in the Malang basin—depths that exceed conventional surface survey and approach or exceed the upper limit of ground-penetrating radar in volcanic sediment. A 51-pair eruption–site literature compilation yields a mean measured depth of 3.41 m over a 760-year horizon, consistent in order of magnitude with the site-level anchors (full dataset intended for supplementary release). Spatial analysis of 666 East Java sites shows near-volcano clustering that reflects survey intensity and monument durability rather than settlement distribution, and is offered as descriptive context not as a test of the taphonomic hypothesis.

The paper closes with a research program (Section 5.7) identifying specific techniques—OSL dating of soil profiles, tephrochronology in soil cores, targeted coring at non-monumental sites—by which these preliminary anchors could be upgraded to validated calibration. What we claim here is that existing evidence is consistent with a substantial Holocene detection horizon in volcanic Java’s interior, sufficient to justify the proposed fieldwork. We do not claim this evidence yet establishes such a horizon.

Keywords: volcanic taphonomy; cumulative sedimentation; detection horizon; East Java; preliminary measurement; research program; geoarchaeology

1 Introduction

In 1803, a Dutch colonial administrator named Nicolaus Engelhard encountered something at the ruins of Candi Singosari that has not received the theoretical attention it deserves. Two enormous stone guardian statues—*Dwarapala*, each 370 cm tall and carved from a single block of andesite—stood with roughly half their bodies below the ground. They had been installed in 1268 CE, during the reign of King Kertanegara of the Singosari kingdom

(Kinney et al., 2003). The ground had risen around them at approximately 3.5 mm per year, driven by 535 years of repeated eruptions from Gunung Kelud, 35 km to the east. Only their scale made them visible at all.

Hindu-Buddhist kingdoms flourished on Java from at least the 4th century CE, yet the material record of the earliest Javanese polities is comparatively sparse. Stone temples of the 8th to 15th centuries survive because they are monumental; organic architecture does not. This paper does not attempt to reconstruct what the pre-monumental record looked like. Here the scope is narrower: to present preliminary measurements of the rate at which cumulative volcanic sedimentation has buried the Javanese landscape, and to evaluate what those measurements imply for the depth at which historical-era deposits now reside. We emphasise the word *preliminary* from the outset: four measurement anchor sites in two volcanic systems are a starting point for a research program, not a validated cross-regional calibration.

The mechanism at work is not the catastrophic burial familiar from Pompeii, Akrotiri, or Cerén—single eruptions sealing settlements in hours (Sigurdsson et al., 1985; Dumas, 1983; Sheets, 1992). It is *cumulative* sedimentation: repeated small eruptions depositing millimeters per event, accumulating to meters over centuries (Sect. 2.3). In Java, every point on the island lies within approximately 27 km of an active vent (van Bemmelen, 1949), placing the entire landmass within the depositional range of at least one volcanic center. The Dwarapala statues provide direct photographic documentation of this process at a known, measurable rate (Fig. 1).

Surface visibility is a recognized problem in archaeological survey (Wandsnider and Camilli, 1992; French, 2003), but its implications for volcanic landscapes in island Southeast Asia have been largely unexplored. Research in the region has focused on monumental stone architecture—the class of evidence most resistant to burial (de Groot, 2009)—producing an inherent circularity: stone temples are found because they protrude through sediment; their distribution is interpreted as reflecting settlement patterns; and the wooden settlements that constituted the overwhelming majority of historical habitation are overlooked entirely.

Dwarapala Guardian Statues at Candi Singosari, Malang
Built ~1268 CE | 370 cm tall | Monolithic andesite | Sedimentation rate: 3.5 mm/yr



Figure 1: Dwarapala guardian statues at Candi Singosari, Malang, East Java. Left: colonial-era photograph (c. 1860) from the Leiden University Library archives, showing the statue partially buried beneath volcanic sediment—only the head, shoulders, and hands remain visible above the accumulated overburden. Right: present-day condition after excavation, revealing the full 370 cm height. The statues were constructed ~1268 CE during the reign of Kertanegara; approximately 185 cm of volcanic overburden accumulated by their rediscovery in 1803 CE, yielding a calibrated sedimentation rate of 3.5 mm/yr (Sect. 3.2).

The empirical foundation of this study is a set of four measurement anchors. We compiled construction dates and measured burial depths for four archaeological sites: the Dwarapala statues at Candi Singosari (~1268 CE, Kelud volcanic system) and three Hindu-Buddhist temples near Merapi in Central Java (Sambisari, Kedulan, and Kimipulan, all 9th century CE). These span roughly 350 km of Java and two geologically distinct volcanic systems. Kelud has produced 37 confirmed eruptions since 1000 CE (Global Volcanism Program, 2024). Merapi is one of the world’s most persistently active stratovolcanoes (Gertisser et al., 2012). The site-level rates fall within a single order of magnitude (2.4–6.2 mm/yr), consistent across both volcanic systems—but with $n = 4$ and shared methodological basis, we treat this as preliminary agreement rather than validated convergence. A rigorous Java-wide calibration requires the further testing outlined in Sect. 5.7.

Those rates are applied to project burial depths for successive historical periods: Kanjuruhan-era (~760 CE) remains in the Malang basin are predicted (at the anchor-site mean rate) to lie at 3.0–7.9 m—depths that exceed the reach of surface survey and

approach or exceed the upper limit of ground-penetrating radar in volcanic sediment (Putra and Setyastuti, 2019; Conyers, 2004). A spatial analysis of 666 known archaeological sites in East Java is included for descriptive context rather than as a test of the anchor-site observations: the distribution is shaped by both survivorship (monumental stone versus organic architecture) and two centuries of survey concentrated on the Singosari–Majapahit heartland, and cannot by itself confirm or refute a taphonomic interpretation.

2 Background

2.1 Java as a volcanic burial zone

Java hosts 45 active volcanoes across approximately 129,000 km²—a volcanic density of 0.35 per 1,000 km², the highest of any major Indonesian island and approximately six times that of Sumatra (0.06/1,000 km²) (van Bemmelen, 1949). The average spacing between volcanic centers is approximately 54 km, meaning that no point on Java lies more than approximately 27 km from the nearest active volcano. Since tephra from VEI 3–4 eruptions can deposit measurable ash at 50–100+ km from the source (Newhall and Self, 1982), the entire island falls within the depositional range of at least one—and typically multiple—volcanic centers. The relevant archaeological question is therefore not whether volcanic sedimentation occurs at a given location, but how much has accumulated since a site was occupied—a function of distance from active vents, local topography, and eruption history.

East Java province alone encompasses seven major active centers: Kelud, Semeru, Arjuno-Welirang, Bromo, Lamongan, Raung, and Ijen. The Brantas River basin—which hosted the Singosari (1222–1293 CE) and Majapahit (1293–~1500 CE) kingdoms—sits in the depositional path of Kelud (35 km east), Arjuno-Welirang (25 km north), and Semeru (60 km southeast). Kelud alone has produced 37 confirmed eruptions since 1000 CE (Global Volcanism Program, 2024; Maeno et al., 2019).

In Central Java, Mount Merapi—one of the world’s most active stratovolcanoes—has buried a cluster of 9th-century Hindu-Buddhist temples to depths of 3–7 m (Lavigne

and Thouret, 2000; Thouret et al., 2000). These buried temples (Sambisari, Kedulan, Kimpulan) were discovered accidentally during modern construction and mining activities, not through systematic archaeological survey.

2.2 The observable archaeological record in East Java

Our compilation of known archaeological sites in East Java (Sect. 3.1) identified 666 unique entries, of which only 391 (59%) have usable spatial coordinates. The dataset is dominated by stone monuments (*candi*) from the Singosari and Majapahit periods (13th–15th centuries CE) (Kinney et al., 2003). Earlier periods are poorly represented: pre-10th century sites constitute less than 5% of the geocoded total.

This temporal distribution reflects two complementary pressures. Survey has historically concentrated on the Singosari-Majapahit heartland around Blitar, Malang, and Mojokerto. Simultaneously, older sites have accumulated more volcanic overburden and are therefore less likely to be exposed at the modern ground surface. The broader question of whether the pre-10th century absence reflects low population, non-monumental material culture, or taphonomic erasure is taken up in a companion synthesis (Amien, in preparation); the present paper addresses only the geophysical rate at which cumulative sedimentation proceeds.

2.3 Volcanic taphonomy

The volcanological literature on archaeological burial is dominated by catastrophic events: Pompeii sealed in eighteen hours, Candi Sambisari buried over roughly a thousand years. These are fundamentally different processes that produce different archaeological signatures.

Catastrophic volcanic burial—a single eruption sealing an entire settlement in hours—is well documented (Sigurdsson et al., 1985; Doumas, 1983; Sheets, 1992) and, precisely because it is dramatic, attracts sustained excavation attention. *Cumulative* volcanic burial, by contrast, generates no single event to excavate. Ash arrives in centimeters per eruption, lahars rework it downslope, and every decade the ground surface is fractionally

higher than the decade before. After five centuries, a surface that was once at ground level lies two meters underground. After fifteen centuries, it may lie at six meters depth. No emergency response is triggered; no excavation follows. The site becomes, gradually and completely, invisible to surface survey.

2.4 A note on comparative context

Two observations from elsewhere in the archipelago inform the calibration without being part of its argument. First, the oldest conventionally recognised Indonesian polity, Kutai Martadipura (~400 CE) in East Kalimantan, is known from inscribed stone pillars found near the present ground surface in a region with zero active volcanoes (Vogel, 1918; Coedès, 1968; Miksic, 2004). Second, three Hindu-Buddhist temples on the Merapi flank (Sambisari, Kedulan, Kimpulan) were discovered beneath 3–7 m of volcanic sediment. These observations bracket the range of outcomes the calibration needs to explain: very shallow burial where no volcanism operates, meter-scale burial where it does. Whether the volcanic/non-volcanic contrast has broader interpretive implications for Indonesian historiography is a question beyond the scope of this paper (see Amien, 2026).

3 Data and Methods

3.1 Archaeological site dataset

We compiled a database of known archaeological sites in East Java province from three complementary sources. First, we queried the OpenStreetMap Overpass API for all features tagged with `historic=*` within the Jawa Timur administrative boundary (bounding box: 6.5–9.0°S, 111.0–115.0°E), yielding 281 geolocated features. Second, we queried the Wikidata SPARQL endpoint for entities with coordinate property P625 within the same bounding box, recovering 16 precisely located sites. Third, we scraped the Indonesian-language Wikipedia article “Daftar candi di Indonesia” for 369 additional entries.

To increase spatial coverage, we geocoded the 369 coordinate-less entries using the OpenStreetMap Nominatim API with progressive query refinement. Of 369 queries, 94

returned valid coordinates within the study area (25.5% success rate). After spatial deduplication within a 100 m radius, the final dataset contains 666 unique site entries, of which 391 have usable coordinates (58.7% geocoding rate). Within the East Java analytical bounds, 383 geocoded sites were used for spatial analysis.

It should be noted that this dataset is heavily biased toward stone monuments (*candi*) that survived volcanic burial due to their monumental scale. Wooden settlements, which likely constituted the vast majority of historical habitation, are systematically absent—precisely the pattern predicted by the taphonomic bias hypothesis.

3.2 The Dwarapala calibration

The primary empirical anchor for the burial depth framework is the pair of Dwarapala guardian statues at Candi Singosari, Malang Regency, East Java. The statues were constructed approximately 1268 CE during the reign of Kertanegara of the Singosari Kingdom and discovered in 1803 CE by the Dutch colonial administrator Nicolaus Engelhard. Each statue stands 370 cm in seated height and weighs approximately 40 tonnes, carved from monolithic andesite. At the time of discovery, approximately half of each statue’s body was below the ground surface (“*separuh tubuh terpendam*”), corresponding to an estimated burial depth of approximately 185 cm accumulated over 535 years (1803 – 1268 = 535 years). The resulting sedimentation rate is:

$$R = \frac{185 \text{ cm}}{535 \text{ yr}} = 0.35 \text{ cm/yr} = 3.5 \text{ mm/yr} \quad (1)$$

This rate is internally consistent with the documented eruption history of Gunung Kelud. The volcano erupted approximately 20 times between 1268 and 1803 CE, and documented VEI 3–4 eruptions deposit 2–20 cm of ash at the distance of Malang (~35 km from the vent). Twenty eruptions at an average of approximately 5 cm per event would account for approximately 100 cm of the 185 cm total burial. The remainder (~85 cm) is attributable to secondary remobilization (lahars, reworked tephra), contributions from the Semeru and Arjuno-Welirang volcanic systems, and non-volcanic aggradation (Global Volcanism Program, 2024; Maeno et al., 2019). The rates reported here represent *total*

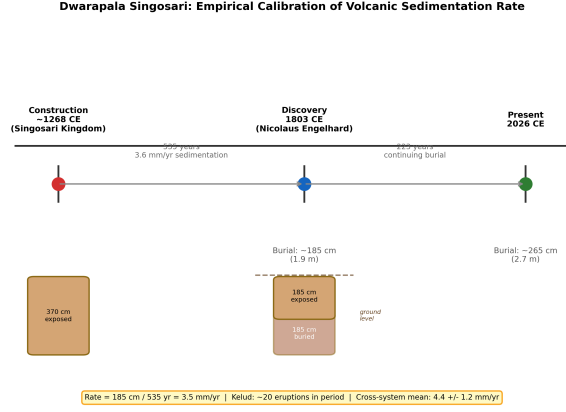


Figure 2: Dwarapala Singosari burial timeline. The statues were constructed ~ 1268 CE at full height (370 cm) and reportedly “half buried” in 1803 CE according to Engelhard’s colonial report, implying approximately 185 cm of volcanic overburden accumulated and a preliminary site-level rate of approximately 3.5 mm/yr. The site-level rates across four measurement anchors range 2.4–6.2 mm/yr.

Table 1: Empirical calibration points for volcanic sedimentation rates at archaeological sites in Java. BPCB-Jatim and BPCB-DIY are Indonesia’s regional heritage conservation agencies (*Balai Pelestarian Cagar Budaya*, Ministry of Education and Culture); burial depths are from their published excavation reports. Sambisari and Kedulan depths are cited in Lavigne and Thouret (2000) and Thouret et al. (2000).

Site	Built (CE)	Discovered	Depth (cm)	System	Rate (mm/yr)	
Dwarapala Singosari	~ 1268	1803	~ 185	Kelud	3.5	BPC
Candi Sambisari	~ 835	1966	500–650	Merapi	4.4–5.7	BPC
Candi Kedulan	~ 869	1993	600–700	Merapi	5.3–6.2	BPC
Candi Kimpulan	~ 900	2009	270–500	Merapi	2.4–4.5	Putra and S

landscape aggradation in volcanic terrain, not pure primary tephra accumulation; for archaeological burial, however, the total aggradation rate is the operationally relevant metric regardless of depositional source.

The Dwarapala burial timeline from construction to present is illustrated in Fig. 2.

3.3 Secondary calibration points

To assess whether the Dwarapala rate is a local anomaly or representative of a Java-wide phenomenon, we compiled three additional calibration points from Central Java’s Merapi volcanic system (Table 1).

Across all four sites, the computed site-level rates span 2.4–6.2 mm/yr. Taking the midpoint of each site’s range (Dwarapala: 3.5; Sambisari: 5.1; Kedulan: 5.8; Kimpulan:

3.5 mm/yr) yields an arithmetic mean across sites of approximately 4.4 mm/yr (the ± 1.2 standard deviation with $n = 4$ carries wide confidence bounds and is reported for description only, not as a precision estimate). The full observed range of 2.4–6.2 mm/yr is used for all subsequent projections. The order-of-magnitude agreement across two volcanic systems—Kelud in East Java and Merapi in Central Java—is consistent with the hypothesis that mm/yr-scale burial of archaeological sites occurs systematically across Java’s volcanic basins (Fig. 3), but the four-site sample, shared methodological basis, and monument-focused selection prevent this paper from establishing that consistency rigorously; see Sect. 5.6 for the constraints and Sect. 5.7 for the proposed research program.

3.4 Literature compilation: 51 eruption–site correlations

As a partial check on the anchor-site rates, we compiled a dataset of 51 documented eruption–archaeological site correlations from across 12 volcanic systems in Island Southeast Asia. Sources were historical and colonial-era records documenting specific volcanic events and their effects on specific known sites; 86% of the evidence was primary (directly observed and recorded, not inferred from proximity). Of the 51 pairs, 24 include physically measured burial depths. The mean measured depth across these 24 sites is 3.41 m (median 2.50 m), order-of-magnitude consistent with the anchor-site mean rate of 4.4 mm/yr applied over a 760-year horizon (expected depth: 3.3 m). While this compilation shares no sites with the four primary anchors, it draws on the same corpus of colonial and volcanological reports and therefore inherits similar selection biases toward monumental and construction-discovered sites; it is best treated as a consistency comparison rather than as formally independent validation. The full dataset (site names, eruption events, dates, measured depths, and primary sources) is intended for supplementary release; aggregate statistics are reported here.

3.5 Burial depth projection model

Using the empirically derived rate range, we estimate expected overburden depth by era according to the linear model:

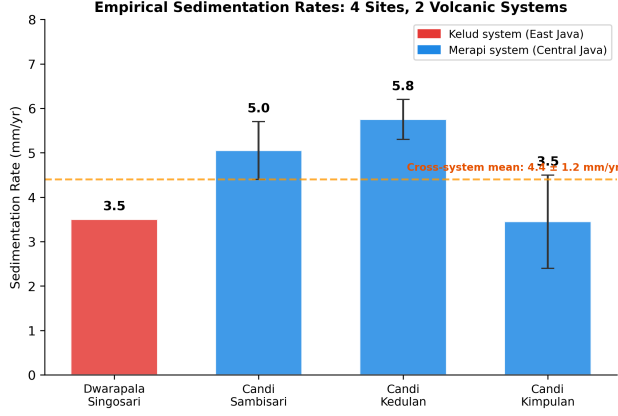


Figure 3: Preliminary site-level sedimentation rates from four archaeological measurement anchors across two volcanic systems. Error bars show the range from reported depth uncertainty. Site-level rates fall within a single order of magnitude (2.4–6.2 mm/yr); the arithmetic mean across sites is approximately 4.4 mm/yr, but with $n = 4$ and shared methodological basis this is not a validated cross-regional calibration.

$$D(\text{era}) = R \times (T_{\text{present}} - T_{\text{era}}) \quad (2)$$

where R ranges from 2.4 to 6.2 mm/yr and $T_{\text{present}} = 2026$ CE. This linear model is a first-order approximation; actual deposition is episodic, with eruption clusters interspersed by quiescent periods and subject to compaction. The wide rate range (2.4–6.2 mm/yr) partially captures this variability. Results are presented in Fig. 4.

3.6 Spatial analysis methods

Two complementary spatial analyses were conducted to examine the relationship between site distribution and volcanic proximity.

In the first analysis, we computed the minimum great-circle distance from each geocoded site to the nearest of seven reference volcanoes. Sites were binned into seven distance bands (0–25, 25–50, 50–75, 75–100, 100–150, 150–200, and 200+ km), and site density was expressed as sites per 1,000 km². Spearman’s rank correlation between distance band midpoint and site density was used to test whether sites are found preferentially near or far from volcanoes.

In the second analysis, we controlled for the effect of terrain suitability on site distribution by constructing a terrain suitability index combining four DEM-derived features:

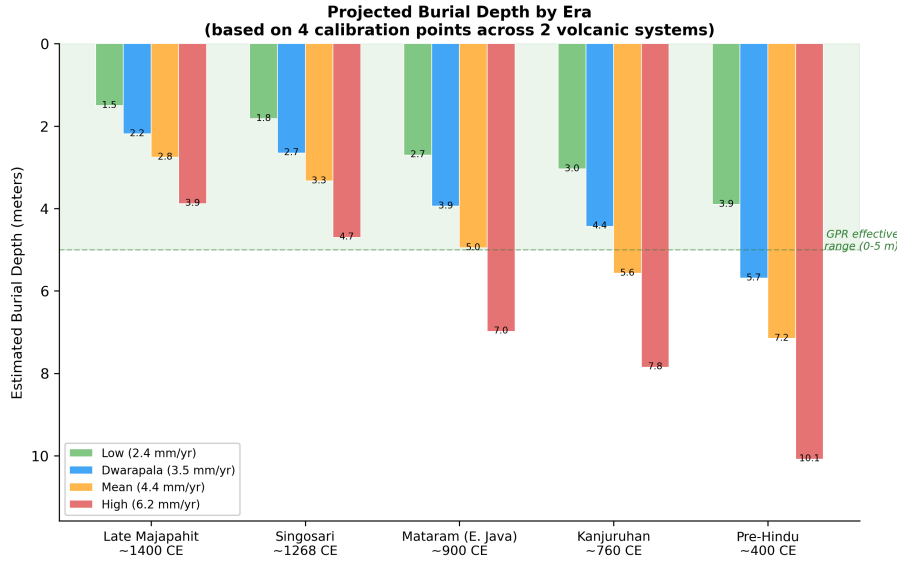


Figure 4: Projected burial depth by historical era using four empirically-derived sedimentation rates. The green band indicates the effective range of ground-penetrating radar in volcanic sediment (0–5 m). Kanjuruhan-era and earlier remains may exceed GPR detection limits.

slope, elevation, topographic wetness index (TWI), and a river proximity proxy. The study area was divided into a 25 km $\times$ 25 km grid (187 cells). For each cell, we computed the residual between observed and terrain-predicted site count. Spearman correlation between residual density and distance to the nearest volcano tests the taphonomic bias hypothesis after controlling for terrain preference.

4 Results

4.1 The Dwarapala measurement anchor

The Dwarapala sedimentation calculation yields a preliminary site-level rate of approximately 3.5 mm/year (Sect. 3.2), conditional on Engelhard’s 1803 “half buried” observation being quantitatively accurate (this is discussed as a limitation in Sect. 5.6). The rate is internally consistent with documented Kelud eruption histories, which account for approximately 100 of the 185 cm total inferred burial through direct tephra deposition alone. The remaining 85 cm is attributable to secondary remobilization and contributions from other volcanic systems.

The Merapi-system anchor sites yield rates of 2.4–6.2 mm/yr, consistently higher than the Dwarapala rate (3.5 mm/yr) from the Kelud system. This difference is physically plausible: Merapi erupts more frequently than Kelud, and the Central Java sites are closer to their volcanic source. The four site-level rates fall within a single order of magnitude—a pattern that is suggestive but does not by itself establish “Java-wide” applicability. With $n = 4$ and a shared methodological basis (depth divided by elapsed time at dated construction), formal calibration requires the additional measurement strategies outlined in Sect. 5.7.

4.2 Literature-compilation comparison

The 51-pair eruption–site correlation dataset is a literature compilation intended as a partial check against the four primary measurement anchors. Of the 51 pairs, 24 include physically measured burial depths, with a mean of 3.41 m (median 2.50 m). Applying the anchor-site mean rate of 4.4 mm/yr over 760 years (the midpoint historical horizon) predicts 3.3 m, which is consistent in order of magnitude with the literature compilation. We emphasise two caveats. First, the 51-pair dataset was compiled from the same corpus of colonial and volcanological reports that document the four anchor sites; the two lines of evidence share selection biases toward monumental and construction-discovered contexts, and therefore are not fully independent in the strict sense. Second, the full 51-pair table (site names, eruption events, dates, measured depths, and primary sources) is intended for supplementary release with this manuscript; the present discussion reports aggregate statistics only. The comparison is consistent with—but cannot by itself confirm—the anchor-site rates as representing a regional baseline rather than a monument-specific artefact.

4.3 Burial depth projections

Table 2 presents key burial depth projections for the Malang basin using the full empirically derived rate range.

These depths exceed standard archaeological surface survey capabilities (0–1 m) and

Table 2: Projected burial depth by era using the preliminary anchor-site rates. **Note: depths are upper bounds (uncompacted projections).** Unconsolidated volcanic sediment compacts by 10–30% at depths exceeding approximately 5 m (Sect. 5.6); realised depths for Kanjuruhan-era and earlier rows are therefore likely 10–30% shallower than shown. The projections assume basin or plain settings with net aggradation; ridge and slope contexts may experience net erosion (Sect. 5.6).

Era	Date (CE)	Elapsed (yr)	Low (2.4)	Dwarapala (3.5)	Mean (4.4)	High (6.4)
Late Majapahit	~1400	626	1.5 m	2.2 m	2.8 m	3.9 m
Singosari	~1268	758	1.8 m	2.7 m	3.3 m	4.7 m
Mataram (E. Java)	~900	1,126	2.7 m	3.9 m	5.0 m	7.0 m
Kanjuruhan	~760	1,266	3.0 m	4.4 m	5.6 m	7.9 m
Pre-Hindu	~400	1,626	3.9 m	5.7 m	7.2 m	10.1 m

approach or exceed the effective range of ground-penetrating radar (2–5 m in volcanic sediment) (Conyers, 2004; Goodman and Piro, 2013). Detection of Kanjuruhan-era or earlier remains will likely require deep GPR, borehole coring, or fortuitous exposure through modern construction.

4.4 Site distribution with respect to volcanic proximity

For descriptive context, we examined how known sites are distributed with respect to distance from active volcanoes. Analysis of 383 geocoded sites yielded a strong negative correlation between site density and distance to the nearest active volcano (Spearman’s $\rho = -0.955$, $p = 0.0008$, $n = 7$ distance bands; Table 3). Known sites cluster near volcanoes, and this is stable under dataset expansion (raw correlation $-0.991 \rightarrow -0.955$ after geocoding 94 additional entries). After controlling for terrain suitability (slope, elevation, topographic wetness index, river proximity) across 187 grid cells, the residual correlation remains negative ($\rho = -0.358$, $p < 0.0001$).

We note that the distribution cannot serve as a test of the calibration framework. The dataset is composed of sites that both survived cumulative burial (favouring monumental stone) and were discovered during two centuries of survey concentrated on the Singosari–Majapahit heartland (Kinney et al., 2003). Several of the deepest finds emerged accidentally during modern construction (Sambisari 1966, Kimpulan 2009, Liangan 2008; Putra and Setyastuti, 2019; Abbas, 2016), indicating that discovery rate is controlled by

Table 3: Site density by distance to nearest active volcano.

Distance Band (km)	Sites	Area (km ²)	Density (/1,000 km ²)
0–25	147	11,343	12.96
25–50	136	18,034	7.54
50–75	37	18,528	2.00
75–100	22	16,373	1.34
100–150	41	28,523	1.44
150–200	0	25,507	0.00
200+	0	73,740	0.00

construction activity more than by systematic archaeological intent. Distance-to-volcano statistics therefore describe where fieldwork has occurred, not where past settlement was concentrated. They are included for transparency, not as inferential evidence. Spatial autocorrelation is also a concern: Moran’s I for volcanic distance across our study grid is 0.937, so simple correlations involving distance are susceptible to inflation and should be interpreted accordingly.

5 Discussion

5.1 Site-level consistency within an order of magnitude

The four measurement anchor sites yield rates that fall within a single order of magnitude (2.4–6.2 mm/yr), spanning two volcanic systems and 350 km of separation. For a process as variable as volcanic sedimentation—driven by eruption frequency, vent distance, wind direction, and local topography—this degree of site-level agreement is suggestive but does not by itself constitute a validated cross-regional calibration. At $n = 4$ the arithmetic mean (approximately 4.4 mm/yr) has wide confidence bounds, and the Kelud and Merapi sites share a common methodological basis (burial depth divided by elapsed time since dated construction), so this is not fully an “independent two-system” comparison in the statistical sense.

The Kelud-system rate (Dwarapala, 3.5 mm/yr) is lower than the Merapi-system rates (central tendency approximately 4.8 mm/yr), which is physically reasonable given that Merapi erupts more frequently. Both systems produce the same order-of-magnitude result,

suggesting—but not establishing—that cumulative volcanic burial operates at broadly comparable rates across Java’s major volcanic basins. Whether this generalises requires the broader sampling strategy outlined in Sect. 5.7.

5.2 Practical implications for fieldwork

The burial depth projections translate directly into survey recommendations that differ significantly from standard practice in Indonesian archaeology. Ground-penetrating radar is effective to approximately 2–5 m in volcanic ash (Conyers, 2004). Late Majapahit-era remains (projected depth 1.5–3.9 m) fall within this detectable range, but Kanjuruhan-era remains (3.0–7.9 m) approach or exceed the practical limits of GPR penetration. Areas with simultaneously high terrain suitability and deep predicted burial represent the highest-priority zones for subsurface investigation, as these are locations where settlement remains are most likely to be preserved yet most likely to remain undetected by surface methods. The pattern of accidental discoveries during modern construction—at Sambisari, Kimpulan, and Liangan—demonstrates that deeply buried sites exist, but systematic subsurface survey targeting such sites has never been conducted in Java.

5.3 What this paper does not claim

We note three boundaries of the present argument. First, the calibration rates are derived from four stone monuments and may not transfer directly to open habitation sites, whose softer material matrix may interact with sediment and erosion differently. The 51-pair eruption–site dataset, which includes non-monumental contexts with a mean measured depth of 3.41 m, is consistent with the four-site calibration and suggests that the monument-to-settlement transfer is reasonable at multi-century timescales; nevertheless, this transfer remains an assumption. Second, a separate regression analysis using three survey-intensity proxies (road accessibility, distance to BPCB heritage conservation offices, and distance to university archaeology departments) shows that volcanic proximity retains a significant negative residual effect on site density (quasi-Poisson likelihood ratio test: $\beta = -0.477$, $p = 0.0015$, $n = 703$ grid cells), suggesting that the volcanic deficit is

not reducible to differential survey effort alone, though this regression is itself subject to the spatial-autocorrelation caveat noted above. Third, we make no claim in this paper about site-type composition (caves versus open-air), population size, cultural complexity, or the broader interpretive question of what a buried pre-Hindu Javanese record might contain. Those questions depend on subsurface investigation that does not yet exist, and on interpretive work that exceeds a geomorphological calibration.

5.4 Limitations

The four measurement anchors, while spanning two volcanic systems and 350 km of Java, constitute a sample whose size and character impose several concrete limits on what this paper can claim. These limits are not cosmetic; they are why we frame the work as preliminary evidence and a research program rather than a validated calibration.

Monument-to-settlement transferability. All four anchor sites are stone monuments (Hindu-Buddhist *candi*), not open habitation sites. Monuments are poor proxies for landscape-scale aggradation in principle: platform construction, foundation excavation, and the vertical mass of the structure itself can trap sediment preferentially or complicate stratigraphic interpretation of the original ground surface. The 51-pair literature compilation includes non-monumental contexts at a mean depth of 3.41 m that is order-of-magnitude consistent with the anchor-site rates, but this partial mitigation does not resolve the fundamental concern. The anchor-site rates should be read as bounds on the aggradation rate experienced by stone monuments, and any extrapolation to open habitation sites is an assumption that requires direct measurement at non-monumental contexts (Sect. 5.7).

Site-level rates also carry measurement uncertainty from imprecise construction dates and variable depth measurements across sources (e.g., Sambisari depths range from “5 m” to “6.5 m” depending on the source).

Dwarapala as anecdotal anchor. The Dwarapala observation—the only Kelud-system anchor—is a qualitative colonial description (“approximately half” of each statue below ground) rather than a primary geoarchaeological measurement. The precise archival

source requires confirmation against the original Engelhard report; we have relied on the consensus secondary-literature reconstruction (Kinney et al., 2003). We have not carried out primary-source archival work at KITLV, the Nationaal Archief, or Leiden University Library where Engelhard’s documents may reside. If the observation proves to reflect sculptural pedestal arrangement, later excavation backfill, or an approximation reported to indicate “most of the statue,” the implied Kelud rate requires substantial revision. The Merapi-system rates are independent of this specific point but share the monument-bias concern above. Future work (Sect. 5.7) proposes specific archival verification.

The sedimentation model is a first-order linear approximation. Actual deposition is episodic, with eruption clusters interspersed by quiescent periods. The wide rate range (2.4–6.2 mm/yr) partially captures this variability at multi-century timescales, but projections to intermediate intervals (50–200 years) should be treated with appropriate uncertainty. Ferring (1986) argues more generally that temporal variability in sedimentation rate rather than the mean rate is the dominant control on archaeological site visibility within a given basin; the multi-century-average framing here is a deliberate simplification, and site-scale visibility at any particular century may differ substantially from the mean-rate projection. Additionally, unconsolidated volcanic sediment compacts over time by 10–30% at depths exceeding roughly 5 m; the depth projections in Table 2 are therefore upper bounds, with realised depths likely somewhat shallower for the oldest eras.

The calibration sites are all located in basin or plain settings (Malang basin, Prambanan plain). On ridges and hillslopes, erosion may equal or exceed deposition, producing net negligible or negative aggradation. The framework’s projections apply to basin contexts; slope locations require separate treatment.

Spatial autocorrelation is strong across the study area: Moran’s $I = 0.937$ for volcanic distance across our grid. Simple correlations involving volcanic distance are therefore susceptible to inflation, and regression-based inferences should be read as suggestive rather than definitive in the absence of formal spatial regression correction.

Finally, rates derive from only two volcanic basins (Kelud and Merapi). Extrapolation to the rest of Java’s volcanic terrain is provisional and requires additional calibration. The

claim that survey history dominates the observed site distribution cannot be quantified precisely without systematic coverage maps, which do not exist for Indonesian archaeological survey.

5.5 A research program for rigorous testing

The preliminary measurement anchors and literature comparison presented above are the starting points, not the conclusions, of a research program. Five specific studies, three of them achievable at modest cost in collaboration with Indonesian institutions already active in the region, would materially upgrade the evidence.

(i) Primary-source archival verification at Dwarapala. The Engelhard 1803 report, associated sketches, and any linked colonial survey documents should be located in archives at KITLV, the Nationaal Archief, and Leiden University Library Special Collections. Quantifying the original “half buried” observation (ideally with measured dimensions rather than qualitative description) would either validate or falsify the Kelud-system anchor.

(ii) Optically Stimulated Luminescence (OSL) dating of soil profiles in non-monumental contexts. Dated soil profiles in the Malang basin and Prambanan plain, collected from settlement-candidate locations rather than adjacent to *candi* foundations, would provide landscape-scale aggradation rates directly applicable to habitation contexts. OSL is well-suited to volcanic sediments where organic preservation is poor; modest cost (~USD 200–500 per sample) means a 20–30 sample programme across two basins is feasible at a single-grant scale.

(iii) Tephrochronology in sediment cores away from monuments. Dated tephra layers in soil cores provide independent chronological anchors and a direct record of cumulative deposition. Merapi and Kelud both have well-constrained Holocene tephra records; cores targeting habitation-candidate basins would yield rates that test the monument-derived anchors.

(iv) Targeted coring and GPR at settlement-candidate sites. Coring at locations identified by independent settlement suitability modelling (see Amien, 2026) would

directly test whether archaeological material survives at the depths projected from the anchor rates. A negative result at a statistically adequate sample (e.g., 20 locations, each reaching the projected pre-400 CE depth) would be material evidence against the detection-horizon interpretation.

(v) Expanded and transparent 51-pair dataset. Full publication of site names, eruption events, dates, measured depths, primary sources, and methodology for the 51-pair compilation as supplementary material would allow other researchers to interrogate selection biases and contribute additional observations. This is a modest-effort deliverable that should accompany an updated version of this paper.

Taken together, these five studies would replace the “preliminary anchors + literature compilation” status of the present evidence with a set of rigorous, non-monumental, and methodologically diverse measurements. At that point, claims of “Java-wide calibration” and specific burial-depth projections for pre-400 CE contexts would be defensible in a way they are not at present.

5.6 Future directions beyond calibration

Once the research program above establishes validated rates, the subsequent practical contribution is in directing fieldwork. A machine learning settlement suitability model that identifies, independently of the burial depth framework, which areas of East Java would have been most attractive to pre-Hindu settlement (Amien, in preparation) can be overlaid on validated rate estimates to produce a prioritised list of subsurface investigation targets—locations where evidence is most likely to exist and least likely to have been found. The buried sites at Sambisari and Kimpulan were found by accident; the framework towards which this paper points provides a basis for finding the next ones deliberately, through targeted GPR survey and deep coring programmes.

6 Conclusions

Four measurement anchors across two volcanic systems and 350 km of separation yield preliminary site-level cumulative aggradation rates that fall within a single order of magnitude (2.4–6.2 mm/yr). With $n = 4$ and a shared methodological basis (burial depth divided by elapsed time since dated construction), we explicitly do *not* claim that this constitutes a validated Java-wide calibration; we claim only that the four sites are consistent with a substantial cumulative-burial regime and that the evidence is sufficient to motivate the research program in Section 5.7. The Dwarapala anchor in particular depends on a qualitative colonial-era observation whose primary-source verification remains outstanding.

The 51-pair eruption–site literature compilation, while order-of-magnitude consistent with the anchor-site rates (mean 3.41 m over a 760-year horizon versus 3.3 m predicted), shares selection biases with the four-site sample and is best treated as a comparison rather than a formally independent validation.

If the preliminary rates are approximately correct, their implications for archaeological detection are material. Kanjuruhan-era remains (~ 760 CE) in the Malang basin would lie at 3.0–7.9 m depth; pre-400 CE remains at 3.9–10.1 m. Both ranges exceed conventional surface survey capability and approach or exceed the upper limit of ground-penetrating radar in volcanic sediment. These projections do not prove that settlement remains exist at these depths in any particular location, and they rest on a rate estimate that itself requires the further testing proposed in Section 5.7. What they do establish is a hypothesis sufficiently well-motivated to warrant rigorous geoarchaeological testing rather than continued assumption either way.

The distribution of known sites provides context but cannot test the framework. The 666-site dataset is composed of monuments that survived burial and that were found by surveys concentrated on the Singosari–Majapahit heartland; any interpretation of the distribution must account for both filters, and the strong spatial autocorrelation (Moran’s $I = 0.937$) in volcanic distance limits what regression-based inference can establish. What can be said without reservation is that the apparent archaeological invisibility of pre-

monumental Javanese settlement is *consistent with* a geomorphologically plausible rate of cumulative burial operating across the historically active volcanic terrain—not that it has been demonstrated to arise from that burial. The paper’s contribution, narrowly, is a set of preliminary measurement anchors and a concrete research programme for testing them.

Data availability

The archaeological site dataset, DEM derivatives, and analysis scripts are available at <https://github.com/neimasilk/volcarch-repo> (to be made public upon acceptance). Raw DEM data from Copernicus GLO-30 are publicly available. A preprint of this manuscript is available at <https://doi.org/10.5281/zenodo.19081502>.

Author contributions

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Competing interests

The authors declare no competing interests.

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